

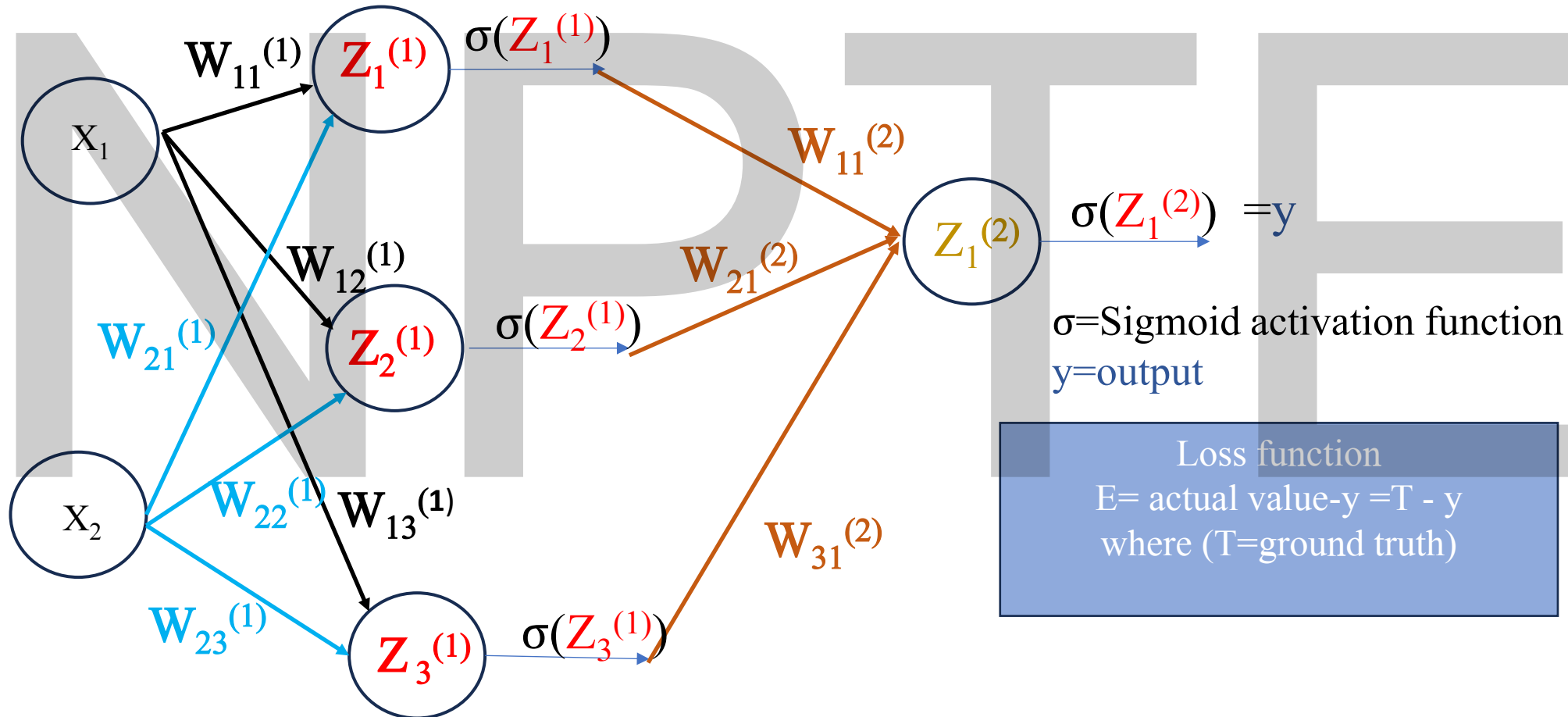
Backpropagation in Neural Network

Backpropagation Algorithm

INPUT LAYER

HIDDEN LAYER

OUTPUT LAYER



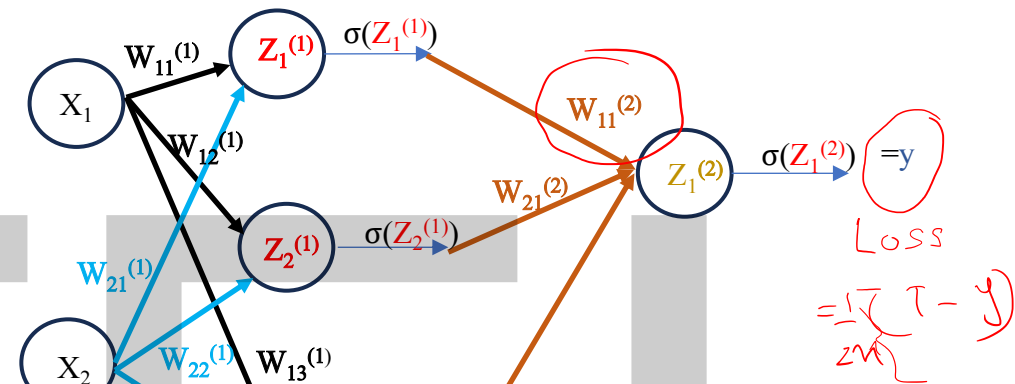
Loss(L) = Mean Squared Error

$$= \frac{1}{2n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$= \frac{1}{2n} \sum_{i=1}^n (T - y)^2 \quad \text{--- ①}$$

According to GD, we need to calculate $\nabla_{\omega} L(\omega, b)$ for weight updation.

$$\nabla_{\omega} L = \frac{\partial L(\omega, b)}{\partial \omega}$$



Let us calculate $\frac{\partial L}{\partial w_{11}^{(2)}}$

$$\frac{\partial L}{\partial w_{11}^{(2)}} = \left[\frac{\partial L}{\partial y} \right] \times \frac{\partial y}{\partial w_{11}^{(2)}} \quad (1.1)$$

from eqn (1)

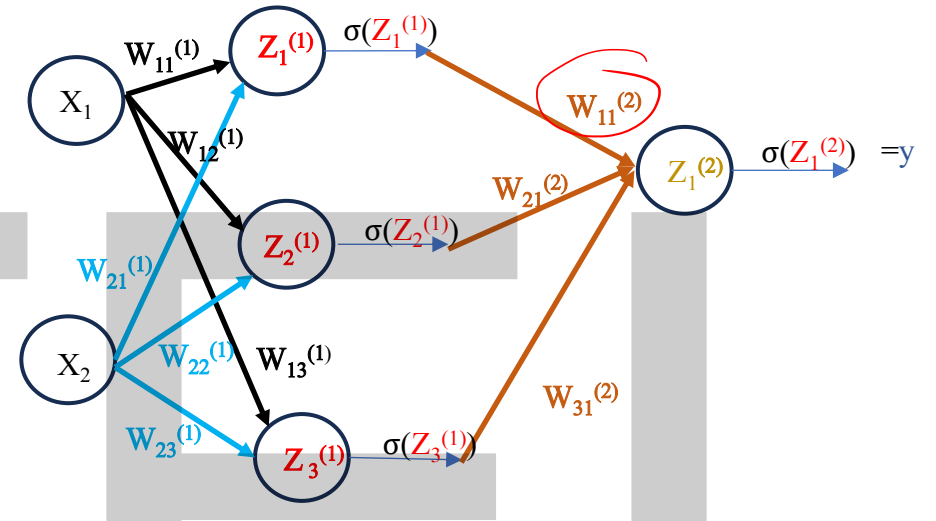
$$\frac{\partial L}{\partial y} = \frac{1}{2n} \sum 2 (T - y) \times (-1) = \frac{1}{n} \sum (y - T) \quad (1.2)$$

$$\frac{\partial y}{\partial w_{11}^{(2)}} = ?$$

$$y = \sigma(z_1^{(2)})$$

$$\sigma(x) = \frac{1}{1 + e^{-x}} = (1 + e^{-x})^{-1}$$

$$\begin{aligned} \sigma'(x) &= -1 (1 + e^{-x})^{-2} \times (e^{-x}) (-1) = \frac{e^{-x}}{(1 + e^{-x})^2} \\ &= \sigma(x) [1 - \sigma(x)] \quad (1.3) \end{aligned}$$



$$y = \sigma(z_1^{(2)})$$

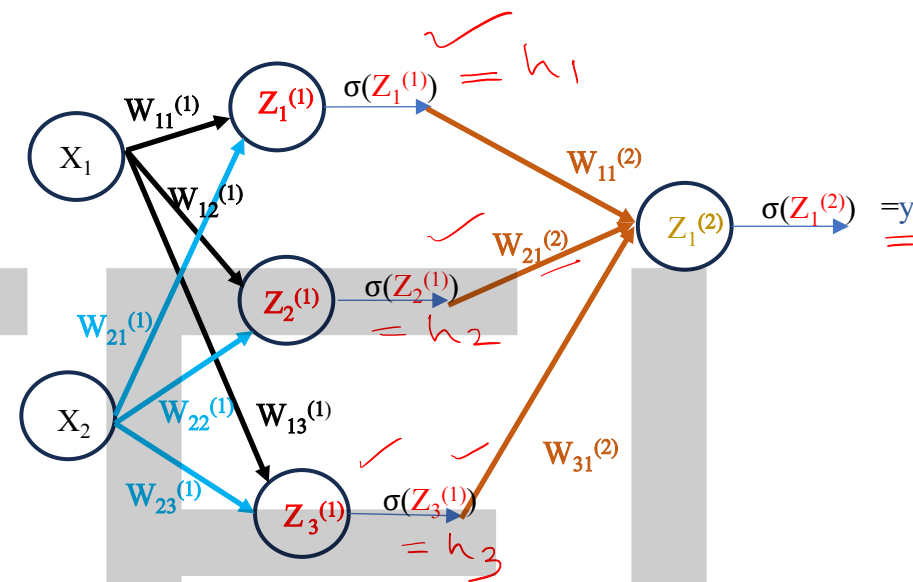
$$\begin{aligned} \therefore \frac{\partial y}{\partial w_{11}^{(2)}} &= \frac{\partial [\sigma(z_1^{(2)})]}{\partial (z_1^{(2)})} \times \frac{\partial (z_1^{(2)})}{\partial w_{11}^{(2)}} \\ &= \sigma(z_1^{(2)}) [1 - \sigma(z_1^{(2)})] \times \frac{\partial (z_1^{(2)})}{\partial w_{11}^{(2)}} \end{aligned}$$

from Eq. (1-3)

Now $\sigma(z_1^{(2)}) = y$

$$\therefore \frac{\partial y}{\partial w_{11}^{(2)}} = y(1-y) \times \frac{\partial (z_1^{(2)})}{\partial w_{11}^{(2)}}$$

$$\text{Now } z_1^{(2)} = h_1 w_{11}^{(2)} + h_2 w_{21}^{(2)} + h_3 w_{31}^{(2)}$$



$$\begin{aligned} h_2 &= \sigma(z_2^{(1)}) \\ h_3 &= \sigma(z_3^{(1)}) \end{aligned}$$

$$\therefore \frac{\partial y}{\partial \omega_{11}^{(2)}} = y(1-y) \times h_1$$

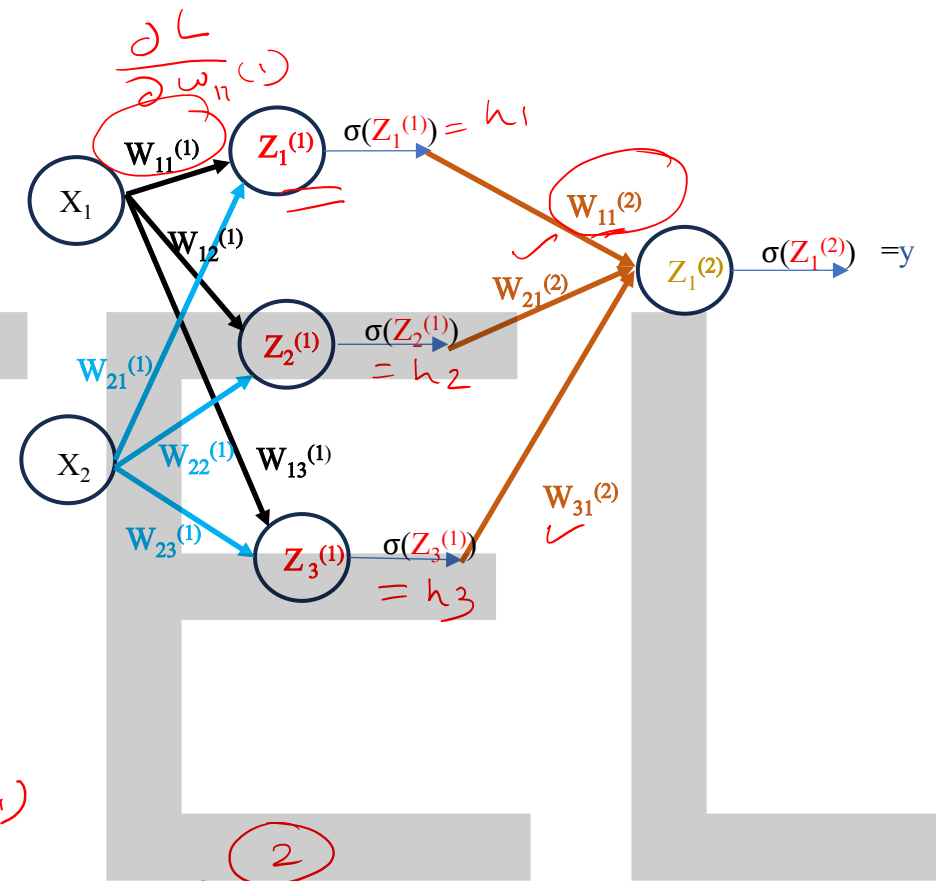
putting all these values in eqn (1.1) we get.

$$\frac{\partial L}{\partial \omega_{11}^{(2)}} = \frac{1}{n} \sum (y \tau) \times y(1-y) \times h_1$$

Now let's calculate $\frac{\partial L}{\partial \omega_{11}^{(1)}}$

$$\frac{\partial L}{\partial \omega_{11}^{(1)}} = \frac{\partial L}{\partial h_1} \times \frac{\partial h_1}{\partial \omega_{11}^{(1)}}$$

$$\begin{aligned} \frac{\partial L}{\partial \omega_{11}^{(1)}} &= \frac{\partial L}{\partial (\sigma(z_1^{(1)}))} \times \frac{\partial (\sigma(z_1^{(1)}))}{\partial z_1^{(1)}} \times \frac{\partial z_1^{(1)}}{\partial \omega_{11}^{(1)}} \\ &= \frac{\partial L}{\partial (\sigma(z_1^{(1)}))} \times \underbrace{\sigma(z_1^{(1)}) [1 - \sigma(z_1^{(1)})]}_{\text{from eqn (1.3)}} \times \frac{\partial z_1^{(1)}}{\partial \omega_{11}^{(1)}} \end{aligned}$$



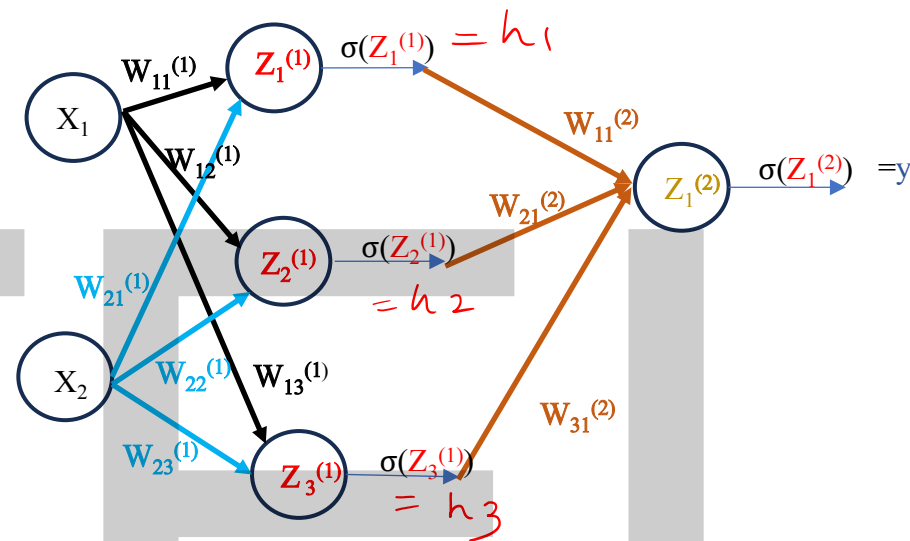
$$\frac{\partial L}{\partial (\sigma(z_1^{(1)}))} \times h_1 [1 - h_1] \times \underline{\underline{x_1}}$$

$$\frac{\partial z_1^{(1)}}{\partial w_{11}^{(1)}} = ?$$

$$z_1^{(1)} = w_{11}^{(1)} x_1 + w_{21}^{(1)} x_2$$

$$\therefore \frac{\partial z_1^{(1)}}{\partial w_{11}^{(1)}} = x_1$$

$$\therefore \frac{\partial L}{\partial w_{11}^{(1)}} = \frac{\partial L}{\partial (\sigma(z_1^{(1)}))} \times h_1 [1 - h_1] \times x_1 \quad \text{--- (3)}$$



$$\frac{\partial L}{\partial (\sigma(z_1^{(1)}))} = \frac{\partial L}{\partial h_1}$$

$$= \frac{\partial L}{\partial y} \times \frac{\partial y}{\partial z_1^{(2)}} \times \frac{\partial z_1^{(2)}}{\partial h_1}$$

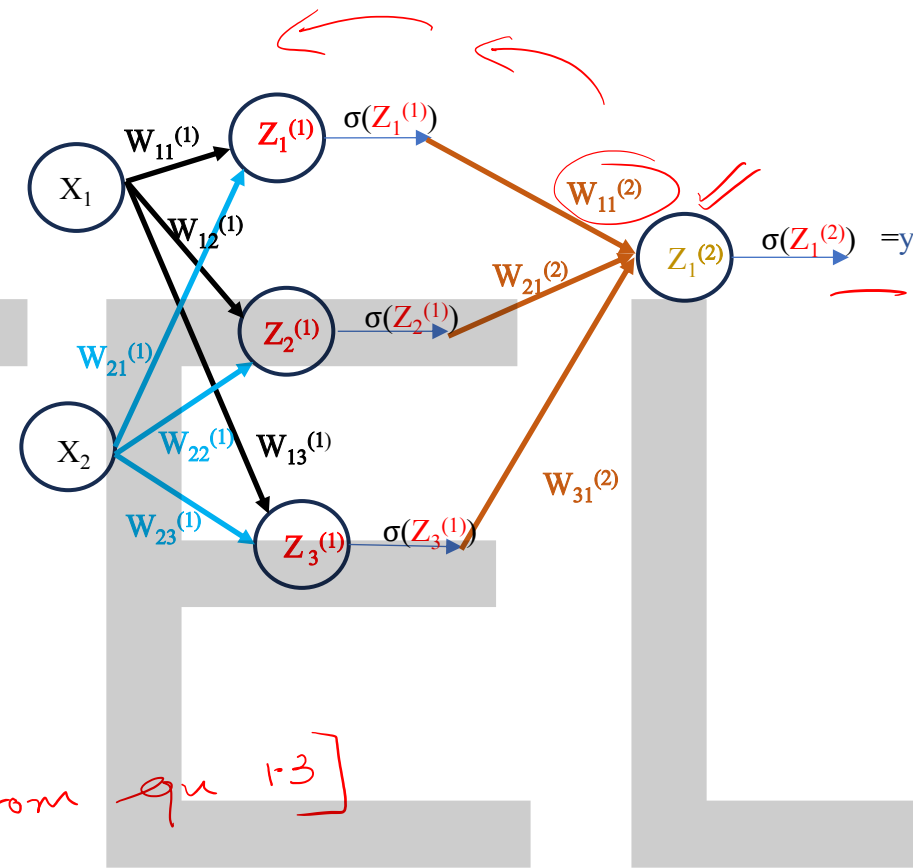
We know $\frac{\partial L}{\partial y} = \frac{1}{n} \sum (y - \hat{y})$ [from eq 1.2]

2nd term:- $\frac{\partial y}{\partial z_1^{(2)}} = \frac{\partial (6 z_1^{(2)})}{\partial z_1^{(2)}}$

$$= 6(z_1^{(2)}) [1 - 6(z_1^{(2)})]$$

$$= y [1 - y]$$

[from eq 1.3]



3rd term:-

$$\frac{\partial z_1^{(2)}}{\partial h_1} = ?$$

$$z_1^{(2)} = h_1 \times w_{11}^{(2)} + h_2 w_{21}^{(2)} + h_3 z_3^{(1)}$$

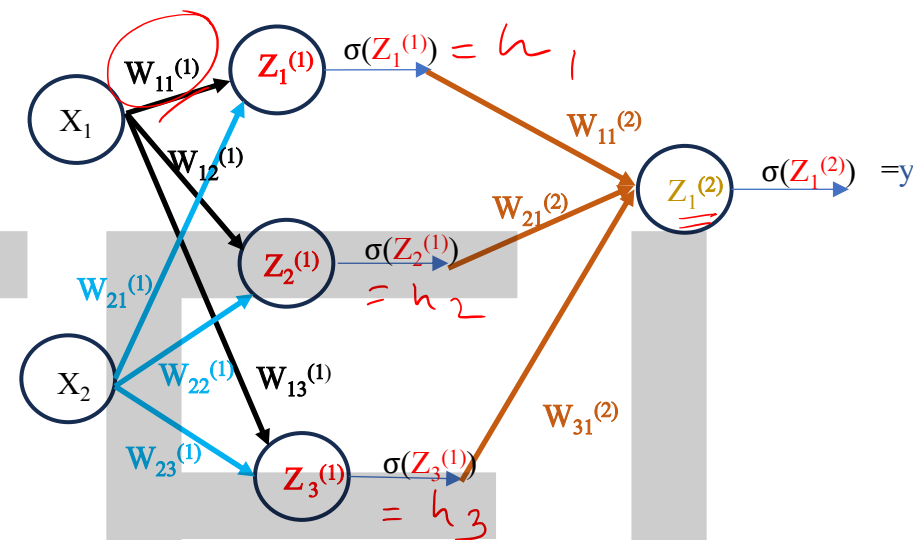
$$\therefore \frac{\partial z_1^{(2)}}{\partial h_1} = w_{11}^{(2)}$$

$$\therefore \frac{\partial L}{\partial h_1} = \frac{1}{n} \sum (y - \hat{y}) \cdot y(1-y) w_{11}^{(2)}$$

putting all these values in eqn (3) we get

$$\int \frac{\partial L}{\partial w_{11}^{(1)}} = \frac{1}{n} \sum (y - \hat{y}) \cdot y(1-y) w_{11}^{(2)} \times h_1 [1 - h_1] \times x_1$$

$\nabla_w L$

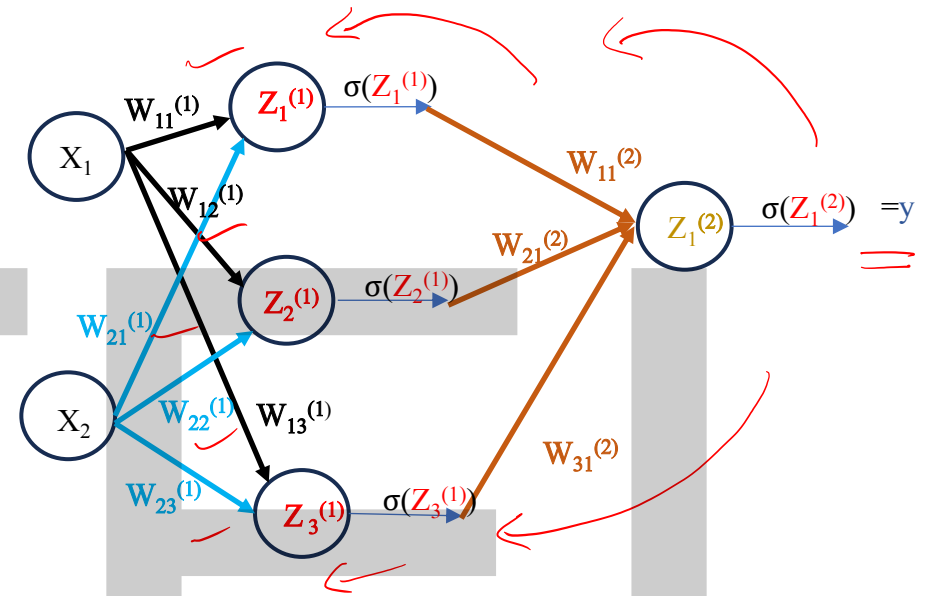


$$W^{(1)} = \begin{bmatrix} W_{11}^{(1)} & W_{12}^{(1)} & W_{13}^{(1)} & \dots \\ W_{21}^{(1)} & W_{22}^{(1)} & W_{23}^{(1)} & \dots \end{bmatrix}$$

Then $\nabla_{\omega} L = \begin{bmatrix} \frac{\partial L}{\partial \omega_{11}^{(1)}} & \frac{\partial L}{\partial \omega_{12}^{(1)}} & \dots \\ \frac{\partial L}{\partial \omega_{21}^{(1)}} & \dots & \dots \end{bmatrix}$

After every iteration, weights will be updated

$$\begin{cases} \omega^{\tau+1} = \omega^{\tau} - \eta \nabla_{\omega} L \rightarrow \text{at } \omega = \omega^{\tau} \\ b^{\tau+1} = b^{\tau} - \eta \nabla_b L \end{cases}$$



NPTEL

Thank you

Next Session:

Variants of Gradient Descent Algorithm and
Momentum Based Optimizers